

- (b) (15 points) What is the minimum costs if the firm want to produce f_0 units of output?

5. (20 points) Ordinary differential equations

- (a) (5 points) Given following equations:

$$\dot{x} = 10x(t), \quad \dot{y} = 5 - 2y(t), \quad \frac{dz}{dt} + 5z(t) = -3$$

provide general solutions and particular solutions with initial conditions

$$x(0) = 0, \quad y(2) = 1, \quad z(20) = 0.$$

Explain your results.

- (b) (5 points) Characterise dynamic equilibria and their stability for equations above.

- (c) (5 points) Consider equation

$$\dot{x} = x^3(t) - x(t).$$

Find its equilibria and characterise their stability.

- (d) (5 points) Sketch at least two solution curves (time paths) for equation from 5c.

This exam has 5 questions, for a total of 100 points.

1. (15 points) **Calculus**

Consider a (piecewise-defined) function:

$$f(x) := \begin{cases} x^5, & x \leq -2 \\ -32, & -5 > x > -2 \\ 5 + \ln x, & x \geq 5. \end{cases}$$

- (a) (3 points) Define regions of monotonicity of $f(x)$: where function is (strictly) increasing/decreasing? Where it is non-strictly increasing/decreasing?
- (b) (3 points) What are discontinuity points of $f(x)$? Compute the derivative of $f(x)$ where it exists.
- (c) (3 points) What are points where $f(x)$ is non-differentiable?
- (d) (4 points) Where is the function $f(x)$ strictly/non-strictly concave/convex?
- (e) (2 points) Sketch the graph of $f(x)$.

2. (20 points) **Implicit functions**

Assume the following system of equations:

$$\begin{aligned} F_1(x, y, u, v) &= x^2 + y^2 - u^2 - v^2 = 0 \\ F_2(x, y, u, v) &= -u + v - xy + 1 = 0 \end{aligned}$$

- (a) (10 points) Prove that this system defines x, y as implicit functions of u, v in the neighbourhood of the point $(2, 1, 1, 2)$
- (b) (10 points) Write down expressions for $\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}$

3. (15 points) **Unconstrained optimization**

Given the profit function

$$\pi(K, L, R) = K^{1/4} L^{1/4} R^{1/4} - 4K - 2L - 4R$$

with $K, L, R > 0$.

- (a) (5 points) Suppose $K = 1$. Show that function $\pi(1, L, R)$ is concave in L, R
- (b) (10 points) Find optimal values of K, L, R which maximize profit function.

4. (30 points) **Constrained optimization**

Let $f(x, y) = \ln x + \ln y$ be production function of a firm with inputs x, y with prices w_1, w_2 (in dollars).

- (a) (15 points) What is the maximum output if firm has I dollars?

1. Assume there are markets for three goods (Q_d^i denotes demand for good i , Q_s^i denotes supply of good i) with interconnected demands and supply. Demand is given for every good by:

$$\begin{aligned} Q_d^1 &:= a - 2P_1 - \frac{1}{5}P_2 + 2P_3, \\ Q_d^2 &:= 30 - P_2 - \frac{2}{5}P_1 + \frac{1}{5}P_3, \\ Q_d^3 &:= 100 - 2P_3 + P_1 + \frac{1}{5}P_2. \end{aligned}$$

Supply is given for every good by:

$$\begin{aligned} Q_s^1 &:= 10 + \frac{1}{5}P_1 + \frac{1}{5}P_2 - P_3, \\ Q_s^2 &:= 2P_2 - \frac{2}{5}P_1 + \frac{1}{5}P_3, \\ Q_s^3 &:= 5 + P_3 + \frac{1}{5}P_1 - \frac{1}{5}P_2. \end{aligned}$$

- (a) (7 points) Compute price elasticities $\epsilon_{Q_d^1, P_1}, \epsilon_{Q_d^1, P_2}, \epsilon_{Q_d^1, P_3}, \epsilon_{Q_d^2, P_1}, \epsilon_{Q_d^2, P_2}, \epsilon_{Q_d^2, P_3}, \epsilon_{Q_d^3, P_1}, \epsilon_{Q_d^3, P_2}, \epsilon_{Q_d^3, P_3}$.
- (b) (8 points) Evaluate these elasticities at a point $P_1 = 1, P_2 = \frac{1}{2}, P_3 = 1$. What can you say about the sign and magnitude of them (elastic/inelastic).
- (c) (5 points) Write down equilibrium conditions for every market. Express the system of equilibrium conditions in the form $A\vec{P} = \vec{d}$, specify $A, \vec{P}, \vec{d}$.
- (d) (6 points) Find equilibrium prices P_1^*, P_2^*, P_3^* . What are conditions on parameter a for all prices to be non-negative?
2. Consider a (piecewise-defined) function:

$$f(x) := \begin{cases} \frac{1}{x}, & x < -1 \\ x^3, & -2 > x \geq -1 \\ e^{\frac{1}{x}}, & 5 > x \geq 2 \\ e^{\frac{1}{5}}, & x \geq 5 \end{cases}$$

- (a) (2 points) Define regions of monotonicity of $f(x)$: where is the function (strictly) increasing/decreasing? Where is it non-strictly increasing/decreasing?
- (b) (2 points) What are discontinuity points of $f(x)$?

- (c) (3 points) What are points where $f(x)$ is non-differentiable? Compute the derivative of $f(x)$ where it exists.
- (d) (4 points) Compute the second derivative. What are regions where function is C^2 ? Where is the function $f(x)$ strictly/non-strictly concave/convex? Are there inflection points?
- (e) (2 points) Sketch the graph of $f(x)$.
3. Given the system of functions (vector-function):
- $$F_1(x, y, u, v) = \ln x + y^2 - \frac{2u}{v}$$
- $$F_2(x, y, u, v) = u^2 - xy + \ln v$$
- (a) (5 points) Prove that this system defines x, y as implicit function of u, v in the neighbourhood of the point $(2, 1, 3, 2)$.
- (b) (9 points) Write down expressions for $\frac{\partial x}{\partial u}$, $\frac{\partial y}{\partial v}$. What can you say about increase/decrease of functions at a point of estimation?
4. Given the profit function
- $$\pi(K, L_1, L_2) = K^{\frac{1}{4}} L_1^{\frac{1}{4}} L_2^{\frac{1}{4}} - 4K - 2L_1 - 4L_2$$
- with $K, L_1, L_2 > 0$, show:
- (a) (2 points) Is the restriction $K, L_1, L_2 > 0$ essential? Why?
- (b) (7 points) Suppose $L_2 = 1$. Show that profit function is concave in K, L_1 .
- (c) (15 points) Find optimal values of K, L_1, L_2 which maximize the profit function.
5. Let $U(x, y) = \ln x + \ln y$ be utility function of a consumer over goods x, y with prices p_1, p_2 .
- (a) (11 points) Assume the consumer has a budget of I dollars at his/her disposal. What is the maximum utility the consumer may achieve with this budget? Prove it is maximum through appropriate second order conditions.

- (b) (12 points) What is the minimum expenditure if the consumer wants to derive U_0 units of utility?

This RESIT exam has 5 questions, for a total of 100 points.

Internal Use Only

1. (15 points) **Calculus**

Consider a (piecewise-defined) function:

$$f(x) := \begin{cases} x^2, & x < 2 \\ 5, & x = 2 \\ 2x, & 5 > x > 2 \\ 10, & 10 > x \geq 5 \\ 10 - 0.1x, & x \geq 10. \end{cases}$$

- (a) (3 points) Define regions of monotonicity of $f(x)$: where function is (strictly) increasing/decreasing? Where it is non-strictly increasing/decreasing?
- (b) (3 points) What are discontinuity points of $f(x)$? Compute the derivative of $f(x)$ where it exists.
- (c) (3 points) What are points where $f(x)$ is non-differentiable?
- (d) (4 points) Where is the function $f(x)$ strictly/non-strictly concave/convex?
- (e) (2 points) Sketch the graph of $f(x)$.

2. (15 points) **Unconstrained Optimization**

Given the profit function

$$\pi(K, L) = AK^{0.3}L^{0.7} - rK - wL$$

with $K, L > 0$, r, w being (constant) prices of capital and labour respectively.

- (a) (2 points) Why is the restriction $K, L > 0$ essential?
- (b) (5 points) Test whether the function is concave for varying K, L . When is it strictly concave?
- (c) (8 points) Find optimal proportion of K, L .

3. (25 points) **Constrained Optimization**

Let $U(x, y) = \ln(x) + \ln(y^2)$ be utility function of a consumer over goods x, y with prices p_1, p_2 .

- (a) (10 points) Assume the consumer has a budget of C dollars at his/her disposal. What is the maximum utility the consumer may achieve with this budget?
- (b) (15 points) What is the minimum expenditure if the consumer wants to derive U_0 units of utility?

4. (25 points) **Integration**

- (a) (7 points) Find following integrals:

$$\int x^5 dx, \int (2x^2 + 3x - 10) dx, \int_0^1 (4e^{-4t}) dt.$$

- (b) (8 points) Using integration by parts, find the following:

$$\int_1^2 5xe^{x-5} dx.$$

- (c) (10 points) Given the marginal costs and marginal revenue functions

$$MC(Q) = 2Q + \frac{1}{2}, \quad MR(Q) = 11 - 2Q$$

and initial conditions $TC(0) = 1, TR(0) = 0$ define the profit function. Which quantity Q provide maximum profit?

5. (20 points) **Ordinary differential equations**

- (a) (5 points) Given following equations:

$$\dot{x} = -x(t), \quad \dot{y} = 4y - 2, \quad 2\frac{dz}{dt} + \frac{1}{10}z(t) = 3$$

provide general solutions and particular solutions with initial conditions

$$x(0) = 1, \quad y(1) = 10, \quad z(2) = 0.$$

Explain your results.

- (b) (5 points) Characterise dynamic equilibria and their stability for equations above.

- (c) (5 points) Consider equation

$$\dot{x} = \frac{x^2(t) - 1}{x^2(t) + 1}.$$

Find its equilibria and characterise their stability.

- (d) (5 points) Sketch at least two solution curves (time paths) for equation from 5c.

This exam has 5 questions, for a total of 100 points.

1. (10 points) **Calculus**

Consider a (piecewise-defined) function:

$$f(x) := \begin{cases} x^4, & x < 0 \\ x^2, & 0 < x \leq 5 \\ 25x, & x \geq 5. \end{cases}$$

- (a) (2 points) Define regions of monotonicity of $f(x)$: where function is (strictly) increasing/decreasing? Where is it non-strictly increasing/decreasing?
- (b) (2 points) What are discontinuity points of $f(x)$?
- (c) (2 points) What are points where $f(x)$ is non-differentiable? Compute derivative where it exists.
- (d) (2 points) Where function $f(x)$ is strictly/non-strictly concave/convex?
- (e) (2 points) Sketch the graph of $f(x)$.

2. (20 points) **Implicit function theorem**

Assume the following system of equations:

$$F_1(x, y, u, v) = x^2 + y - u^2 - v^2 = 0$$

$$F_2(x, y, u, v) = -u + v - xy + 1 = 0.$$

- (a) (10 points) Prove that this system defines x, y as implicit functions of u, v in the neighbourhood of the point $(2, 1, 1, 2)$.
- (b) (10 points) Write down expressions for $\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}$.

3. (20 points) **Unconstrained optimization**

Given the profit function

$$\pi(K, L, R) = K^{1/4} L^{1/4} R^{1/4} - 4K - 2L - 4R$$

with $K, L, R > 0$.

- (a) (5 points) Suppose $K = 1$. Show that function $\pi(1, L, R)$ is concave in L, R .
- (b) (15 points) Find optimal values of K, L, R which maximize profit function.

4. (30 points) **Constrained optimization**

Let $f(x, y) = \ln x + \ln y$ be production function of a firm with inputs x, y with prices w_1, w_2 (in dollars).

- (a) (15 points) What is the maximum output if firm has I dollars?
- (b) (15 points) What is the minimum costs if the firm want to produce f_0 units of output?

5. (20 points) **Ordinary differential equations**

- (a) (5 points) Given the following equations:

$$\dot{x} = 1 - 2x(t), \quad \dot{y} = 4y(t), \quad 3\frac{dz}{dt} - 2z(t) + 2 = 0.$$

Provide general solutions and particular solutions with initial conditions

$$x(0) = 2, \quad y(1) = 10, \quad z(2) = 0.$$

Explain your results.

- (b) (5 points) Characterise dynamic equilibria and their stability for equations above.
- (c) (10 points) Consider equation

$$\dot{x} = x^2(t) + x(t) - 1.$$

Find its equilibria and characterise their stability. Sketch two solution curves.

This RESIT exam has 5 questions, for a total of 100 points.

1. Assume given two non-zero vectors

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \dots \\ x_n \end{bmatrix}, \vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \dots \\ y_n \end{bmatrix}$$

- (a) (4 points) Express $x'y$ and xy' . Does it hold that $y'x = x'y$? If so why?
- (b) (2 points) Show that $x \cdot y = x'y$. Explain your result
- (c) (3 points) Now let

$$x = \begin{bmatrix} 1 \\ 6 \\ 3 \end{bmatrix}, y = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, z = \begin{bmatrix} 5 \\ 1 \\ 7 \end{bmatrix}$$

Compute all pairwise *inner* products $x \cdot y$, $x \cdot z$, $y \cdot z$. Show that $x \cdot z = z \cdot x$

- (d) (5 points) Compute $3x$, $2z+x$, $x-2y+3z$. In Euclidean space $\mathbf{R}^3$ sketch operations and resulting vectors.
- (e) (3 points) What is the dimension of vector space spanned by these vectors x, y, z ? Could you suggest a basis for this vector space?
- (f) (4 points) Come up with the distance function for this space and verify all distance function's properties for it. Compute distance $d(x, y)$, $d(x, z)$, $d(y, z)$ for vectors above.

2. Consider a function:

$$f(x) = \begin{cases} 2, & x < 2 \\ (x-2)^3, & 5 > x \geq 2 \\ \frac{50}{x^2}, & 10 > x \geq 5 \\ \frac{1}{2}, & x \geq 10 \end{cases}$$

- (a) (6 points) Is this function continuous everywhere? If not, what are discontinuity points?
- (b) (4 points) Is it differentiable everywhere? If not, what are points of non-differentiability? Are they the same as discontinuity points?
- (c) (5 points) Compute the derivative of $f(x)$ at all differentiable segments
- (d) (5 points) Using information from above, sketch the graph of $f(x)$

3. Given the system of functions:

$$F_1(x, y, u, v) = \ln x + y^2 - \frac{1}{v}$$

$$F_2(x, y, u, v) = u^2 + xy - \ln v$$

- (a) (5 points) Prove that this system defines x, y as implicit function of u, v in the neighbourhood of the point $(2, 0, 3, 2)$.
- (b) (9 points) Write down expressions for $\frac{\partial x}{\partial u}, \frac{\partial y}{\partial v}$. What can you say about increase/decrease of functions at a point of estimation?
4. Given the cost function of a multi-product firm
- $$C(Q_1, Q_2) := 2Q_1^2 + 2Q_2^2 + 10Q_1 + Q_2 - 5Q_1Q_2 + 100$$
- with $Q_1, Q_2 \geq 0$.
- (a) (2 points) Determine variable costs part and fixed costs part
- (b) (5 points) Suppose $Q_2 = 1$. Show that cost function is convex in Q_1 .
- (c) (15 points) Find optimal values of Q_1, Q_2 which minimise cost function.
5. Let $U(x, y) = (x + 1)^2(y - 3)^2$ be utility function of a consumer over goods x, y with prices p_1, p_2 .
- (a) (3 points) What are feasible values for x, y , assuming utility has to be non-negative?
- (b) (11 points) Assume the consumer has a budget of I dollars at his/her disposal. Find critical values x^*, y^* satisfying first-order conditions for constrained optimum.
- (c) (9 points) Check whether sufficient condition for x^*, y^* from above to be the maximum point, if $I = 10$ is satisfied.

This exam has 5 questions, for a total of 100 points.